

DRUG REPURPOSING DEMO

Query : 'migraine' → migraine disorder

Model : HGT

Scoring 100 drug candidates ...

Computing disease similarity ...

OUTPUT 1 — TOP 5 CANDIDATE DRUGS

Disease : migraine disorder | Model : HGT

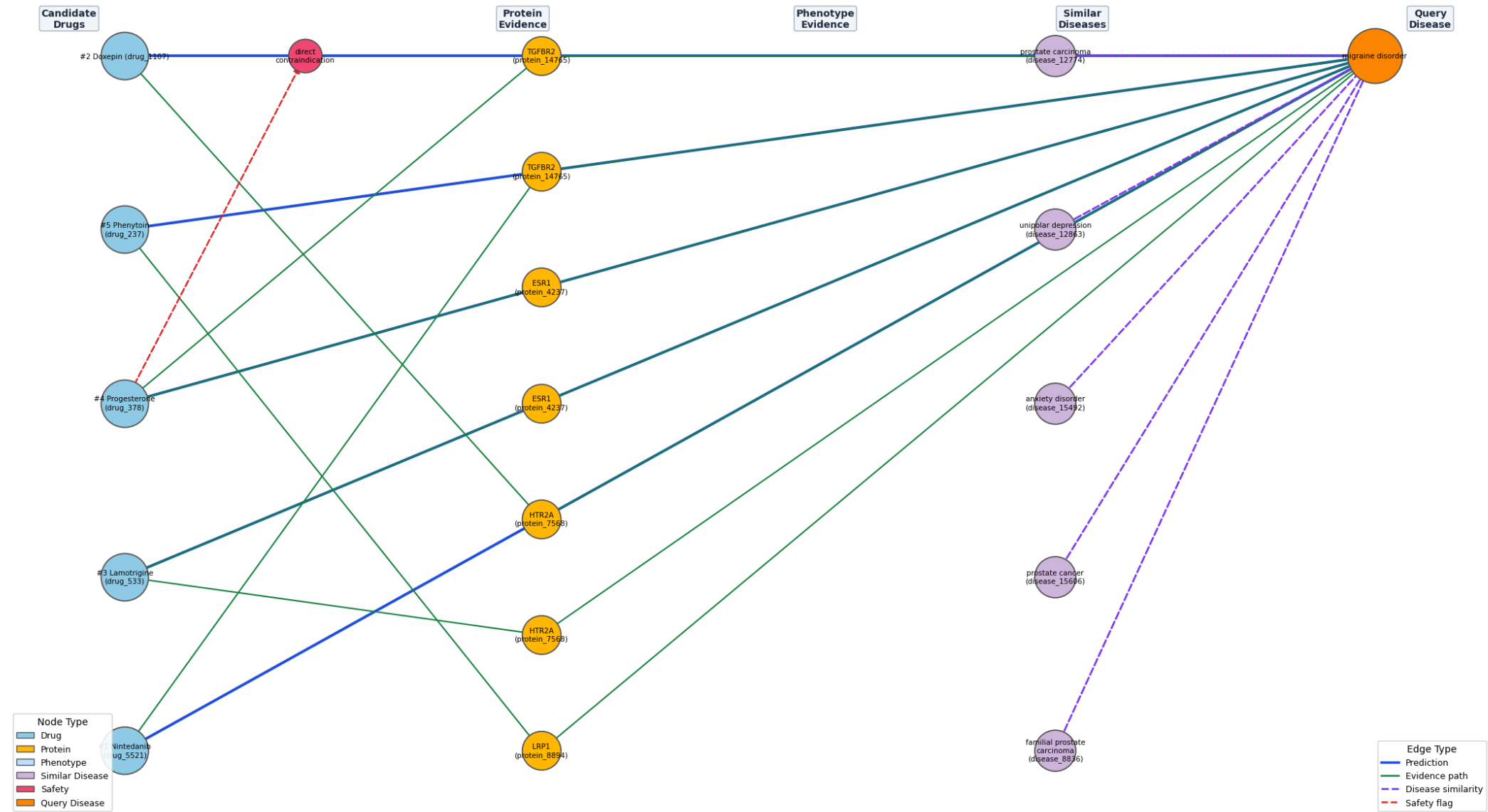
Drug		Model Score	Sim. Support	Rerank Score	Risk	Evidence	Safety Flags	Known Treatment For
Rank								
1	Nintedanib (drug_5521)	0.565758	0.236155	0.520206	low	3	0	acute interstitial pneumonia, idiopathic interstitial pneumonia, non-small cell lung carcinoma (disease) (+2 more)
2	Doxepin (drug_1107)	0.565872	0.230991	0.478978	low	2	0	dermatitis, atopic, insomnia (disease), atopic eczema (+3 more)
3	Lamotrigine (drug_533)	0.565785	0.000000	0.471182	low	3	0	visual epilepsy, cutis verticis gyrata, epilepsy
4	Progesterone (drug_378)	0.565854	0.000000	0.396219	high	3	1	endometriosis of uterus, postmenopausal atrophic vaginitis, endometrium neoplasm (+3 more)
5	Phenytoin (drug_237)	0.565896	0.000000	0.361243	low	1	0	visual epilepsy, Lennox-Gastaut syndrome, status epilepticus (+3 more)

OUTPUT 2 — TOP 10 SIMILAR DISEASES

Similar Disease		Similarity Score	Embedding Cos	Shared Proteins	Shared Effects	Shared Drugs
Rank						
1	familial prostate carcinoma (disease_8836)	0.638075	0.967754	4	0	0
2	prostate carcinoma (disease_12774)	0.612707	0.986160	3	0	0
3	prostate cancer (disease_15606)	0.611220	0.977666	3	0	0
4	anxiety disorder (disease_15492)	0.603288	0.988454	3	0	0
5	unipolar depression (disease_12863)	0.601884	0.994502	3	0	0
6	migraine with or without aura, susceptibility to (disease_1476)	0.583472	0.940466	3	0	9
7	lung cancer (disease_16106)	0.575666	0.989300	2	0	0
8	schizophrenia (disease_12715)	0.574439	0.784823	3	0	0
9	colorectal cancer (disease_13149)	0.573458	0.989514	2	0	0
10	bipolar disorder (disease_12626)	0.563081	0.997556	2	0	0

OUTPUT 3 — EVIDENCE & SIMILARITY SUBGRAPH

Drug Repurposing Evidence Graph — migraine disorder (model: HGT)
Drugs · Protein Evidence · Phenotypes · Similar Diseases · Safety



Subgraph saved → /mnt/data/demo_subgraph.png

OUTPUT 4 — RAG EXPLANATION & SUMMARY

Disease query : migraine disorder
Model : HGT

SIMILAR DISEASES USED AS CONTEXT

Diseases most biologically similar to the query were identified using:
embedding cosine similarity · disease ontology edges · shared proteins
PPI-expanded protein bridges · phenotype overlap · shared therapeutics

- familial prostate carcinoma (disease_8836) (similarity=0.638, embedding=0.968, shared proteins=4, shared effects=0)
- prostate carcinoma (disease_12774) (similarity=0.613, embedding=0.986, shared proteins=3, shared effects=0)
- prostate cancer (disease_15606) (similarity=0.611, embedding=0.978, shared proteins=3, shared effects=0)
- anxiety disorder (disease_15492) (similarity=0.603, embedding=0.988, shared proteins=3, shared effects=0)
- unipolar depression (disease_12863) (similarity=0.602, embedding=0.995, shared proteins=3, shared effects=0)

TOP 5 CANDIDATE DRUGS

Rank 1 – Nintedanib (drug_5521)

Currently known to treat : acute interstitial pneumonia, idiopathic interstitial pneumonia, non-small cell lung carcinoma (disease) (+2 more)
Raw model score : 0.566
Similar disease support : 0.236
Final rerank score : 0.520
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Protein interaction bridge: Nintedanib (drug_5521) --[drug_target_protein]--> FGFR1 (protein_4564) --[protein_interacts_with_protein]--> TGFBR2 (protein_14765) <--[disease_associated_with_protein]--> idiopathic interstitial pneumonia (disease_12863)
- Protein interaction bridge: Nintedanib (drug_5521) --[drug_target_protein]--> PDGFRA (protein_10515) --[protein_interacts_with_protein]--> TGFBR2 (protein_14765) <--[disease_associated_with_protein]--> idiopathic interstitial pneumonia (disease_12863)
- Protein interaction bridge: Nintedanib (drug_5521) --[drug_target_protein]--> SRC (protein_14428) --[protein_interacts_with_protein]--> TGFBR2 (protein_14765) <--[disease_associated_with_protein]--> idiopathic interstitial pneumonia (disease_12863)

Evidence profile:

- Shared protein bridges : 0
- Protein interaction bridges : 3
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

lung cancer (disease_16106)

Rank 2 – Doxepin (drug_1107)

Currently known to treat : dermatitis, atopic, insomnia (disease), atopic eczema (+3 more)
Raw model score : 0.566
Similar disease support : 0.231
Final rerank score : 0.479
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

- Shared protein target: Doxepin (drug_1107) --[drug_target_protein]--> HTR2A (protein_7568) <--[disease_associated_with_protein]--> migraine disorder (disease_12873)
- Protein interaction bridge: Doxepin (drug_1107) --[drug_target_protein]--> HTR1A (protein_7563) --[protein_interacts_with_protein]--> HTR2A (protein_7568) <--[disease_associated_with_protein]--> migraine disorder (disease_12873)

Evidence profile:

- Shared protein bridges : 1
- Protein interaction bridges : 1
- Disease-to-disease transfer : 0
- Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:
bipolar disorder (disease_12626)

Rank 3 – Lamotrigine (drug_533)

Currently known to treat : visual epilepsy, cutis verticis gyrata, epilepsy
Raw model score : 0.566
Similar disease support : 0.000
Final rerank score : 0.471
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:
• Shared protein target: Lamotrigine (drug_533) --[drug_target_protein]--> HTR2A (protein_7568) <--[disease_associated_with_protein]-- migraine disorder (disease_12873)
• Protein interaction bridge: Lamotrigine (drug_533) --[drug_target_protein]--> ADRB1 (protein_3248) --[protein_interacts_with_protein]--> ESR1 (protein_4237) <--[disease_associated_with_protein]-- migraine disorder (disease_12873)
• Protein interaction bridge: Lamotrigine (drug_533) --[drug_target_protein]--> DRD2 (protein_3817) --[protein_interacts_with_protein]--> HTR2A (protein_7568) <--[disease_associated_with_protein]-- migraine disorder (disease_12873)

Evidence profile:
• Shared protein bridges : 1
• Protein interaction bridges : 2
• Disease-to-disease transfer : 0
• Phenotype / effect overlap : 0

Safety signals:
• No direct contraindication or adverse overlap signal found.

Support from similar diseases:
No strong support identified from similar diseases.

Rank 4 – Progesterone (drug_378)

Currently known to treat : endometriosis of uterus, postmenopausal atrophic vaginitis, endometrium neoplasm (+3 more)
Raw model score : 0.566
Similar disease support : 0.000
Final rerank score : 0.396
Risk assessment : high observed graph-based safety concern

Mechanistic evidence:
• Shared protein target: Progesterone (drug_378) --[drug_target_protein]--> ESR1 (protein_4237) <--[disease_associated_with_protein]-- migraine disorder (disease_12873)
• Protein interaction bridge: Progesterone (drug_378) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> ESR1 (protein_4237) <--[disease_associated_with_protein]-- migraine disorder (disease_12873)
• Protein interaction bridge: Progesterone (drug_378) --[drug_target_protein]--> AR (protein_8143) --[protein_interacts_with_protein]--> TGFBR2 (protein_14765) <--[disease_associated_with_protein]-- migraine disorder (disease_12873)

Evidence profile:
• Shared protein bridges : 1
• Protein interaction bridges : 2
• Disease-to-disease transfer : 0
• Phenotype / effect overlap : 0

Safety signals:
• Direct safety warning: Direct contraindication: Progesterone (drug_378) --[contraindication]-- migraine disorder (disease_12873)

Support from similar diseases:
No strong support identified from similar diseases.

Rank 5 – Phenytoin (drug_237)

Currently known to treat : visual epilepsy, Lennox-Gastaut syndrome, status epilepticus (+3 more)
Raw model score : 0.566
Similar disease support : 0.000
Final rerank score : 0.361
Risk assessment : low observed graph-based safety concern

Mechanistic evidence:

• Protein interaction bridge: Phenytoin (drug_237) --[drug_target_protein]--> CACNA1A (protein_15376) --[protein_interacts_with_protein]--> LRP1 (protein_8894) <--[disease_associated_with_p

- Evidence profile:
- Shared protein bridges : 0
 - Protein interaction bridges : 1
 - Disease-to-disease transfer : 0
 - Phenotype / effect overlap : 0

Safety signals:

- No direct contraindication or adverse overlap signal found.

Support from similar diseases:

No strong support identified from similar diseases.

INTERPRETATION NOTE

These are graph-based research hypotheses, not clinical prescriptions.
The strongest candidates combine: high model score + strong mechanistic evidence (especially shared protein bridges) + support from biologically similar diseases + no direct contraindication signals.

Always validate predictions against current clinical literature before drawing any experimental or clinical conclusions.

Structured RAG report table:

rank	drug	known_treatment_for	model_score	similar_disease_support	rerank_score	risk	evidence_summary	safety_summary
1	Nintedanib (drug_5521)	acute interstitial pneumonia, idiopathic interstitial pneumonia, non-small cell lung carcinoma (disease) (+2 more)	0.565758	0.236155	0.520206	low	Protein interaction bridge: Nintedanib (drug_5521) --[drug_target_protein]--> FGFR1 (protein_4564) --[protein_intera...	No direct contraindication or adverse overlap signal found.
2	Doxepin (drug_1107)	dermatitis, atopic, insomnia (disease), atopic eczema (+3 more)	0.565872	0.230991	0.478978	low	Shared protein target: Doxepin (drug_1107) --[drug_target_protein]--> HTR2A (protein_7568) <--[disease_associated_wi...	No direct contraindication or adverse overlap signal found.
3	Lamotrigine (drug_533)	visual epilepsy, cutis verticis gyrata, epilepsy	0.565785	0.000000	0.471182	low	Shared protein target: Lamotrigine (drug_533) --[drug_target_protein]--> HTR2A (protein_7568) <--[disease_associated...	No direct contraindication or adverse overlap signal found.
4	Progesterone (drug_378)	endometriosis of uterus, postmenopausal atrophic vaginitis, endometrium neoplasm (+3 more)	0.565854	0.000000	0.396219	high	Shared protein target: Progesterone (drug_378) --[drug_target_protein]--> ESR1 (protein_4237) <--[disease_associated...	Direct safety warning: Direct contraindication: Progesterone (drug_378) -[contraindication]-- migraine disorder (di...
5	Phenytoin (drug_237)	visual epilepsy, Lennox-Gastaut syndrome, status epilepticus (+3 more)	0.565896	0.000000	0.361243	low	Protein interaction bridge: Phenytoin (drug_237) --[drug_target_protein]--> CACNA1A (protein_15376) --[protein_inter...	No direct contraindication or adverse overlap signal found.